

Chapter 1

Literature Review

This chapter aims to add context to the entire project by exploring the existing literature on RADAR systems, the OFDM modulation scheme, and the combination of the two within a single system in order to achieve the objective of object classification. In order to get to a level where the topic of object classification can be discussed, one must first delve into the fundamental operation of RADAR and OFDM based systems. Once one has a strong understanding of these basics, the combination of the two can be discussed in regard to object classification.

1.1 RADAR

1.1.1 Brief RADAR History

The basic idea of RADAR was experimented with in the classical electromagnetic radiation experiments of the 1880s performed by Heinrich Hertz [1]. These experiments set out to prove the previous theoretical work done by James Clerk Maxwell. Maxwell came up with a formula that described electromagnetic fields, and he hypothesized that this formula would apply to both light and radio waves since they were both examples of electromagnetic radiation, just at very different frequencies. The work done by Maxwell led to the discovery that electromagnetic waves can be reflected by metallic objects and refracted by a dielectric medium, just like light can [1]. This work did not go unnoticed, and a patent for an ‘an obstacle detector and ship navigation device’ was issued in 1904 to Christian Hülsmeyer, a German engineer [1]. Work on RADAR then slowed down due to a lack of interest from governmental authorities until the early 1930s, when the development of long-range bombers prompted the major countries of the world to develop technologies which could detect these hostile aircraft.

1.1.2 RADAR Basics

The fundamental operation of a simple single pulse RADAR system relies on three physical fundamental properties of the universe [2] which are listed below:

1. **Reflection of electromagnetic waves:** This rule describes how a radio wave will be reflected back towards a transmitter of said radio wave if the object is placed directly in front of the transmitter. This reflection indicates that the object is in front of the transmitter.
2. **Electromagnetic energy travels through air at a constant speed:** This rule describes how the electromagnetic waves transmitted by an antenna propagate through the transmission medium, usually air for our earth’s use cases, at roughly the speed of light, c . This velocity c equates to $299,792,458 \text{ m.s}^{-1}$
3. **Electromagnetic energy normally travels through space in a straight line:** This describes how the energy transmitted by an antenna in a radar system propagates through space in a

straight line. This is important because it means that the energy of the RADAR system can be directed in a specific direction by the antenna system, which means that the reflecting object's direction (azimuth and elevation) can be determined.

Using the above-mentioned physical properties, the distance, azimuth, and elevation of the object that is being looked at by the radar system can be determined.

Since RADAR utilizes the electromagnetic spectrum, there are distinct advantages concerning the use of RADAR systems [2]. The main advantage is that, unlike humans, RADAR systems can operate day and night, 24/7, over far distances, in good or bad weather. Compared to human eyes, this is much better. The above example of a RADAR system is that of a simple single pulse RADAR. In the next section I want to outline the other type of RADAR system that I might use in my OFDM system, a pulsed RADAR system.

1.2 OFDM

1.2.1 Brief OFDM History

The idea of OFDM transmission techniques has been around for many years, however, it was never implemented in communication channels until the appearance of mass market applications coincided with the availability of efficient software and electronic technologies [3]. Before OFDM however, there was [Frequency Division Multiplexing \(FDM\)](#).

FDM

FDM system innovation was built out of necessity by telegraph companies hoping to increase the amount of users that could utilize their transmission lines so as to increase profits from their systems [3]. Transmission companies needed a method to have multiple messages being sent on the same transmission line at one time. Other methods had been used such as [Time Division Multiplexing \(TDM\)](#), and [Time Division Multiple Access \(TDMA\)](#). These methods were however limited by burst speed which was itself limited by inter-symbol interference. Around the 1920s, FDM became the main carrier system for telephone companies. The first major company to deploy the system was [American Telephone and Telegraph Company \(AT&T\)](#). This first system used 8kHz subchannel spacing with 4kHz channel size. This form of transmission system later developed into systems that had more sub carrier channels with less subcarrier spacing. There were however some drawbacks to FDM, which are highlighted below:

1. **Waste of frequency:** Due to the large subcarrier spacing, FDM wasted a lot of precious frequency spectrum to these so-called 'guard' bands. As the frequency spectrum became more crowded throughout the years, these guards band became a liability because spectrum access became so crowded.
2. **Using different modulators for the various sub-channels:** In the old systems, each sub-channel had a separate modulator in the up-mixing process. This multiplication process became tedious once there were lots of sub-carriers.

To compensate for the waste of frequency issue, the concept of OFDM was introduced, where each subcarrier overlapped except they had non-interfering spectra [3].

1.2.2 OFDM Basics

Orthogonal Frequency Division Multiplexing (OFDM) is a method of data transmission that breaks information into bit chunks that are sent in parallel over multiple channels [4]. The OFDM transmission scheme builds upon FDM where multiple channels are created over a specific bandwidth, and bits of information are sent on each channel in parallel. The main difference between FDM and OFDM is therefore the orthogonality of the OFDM signal. Pictured below in figure 1.1 is an OFDM, and FDM based waveform. As can be seen, all the peaks of the OFDM waveform have zero interference with any other part of the signal, whereas, with the FDM waveform, the peaks have significant interference from the other channels since the channels are not spaced orthogonally in regard to each other.

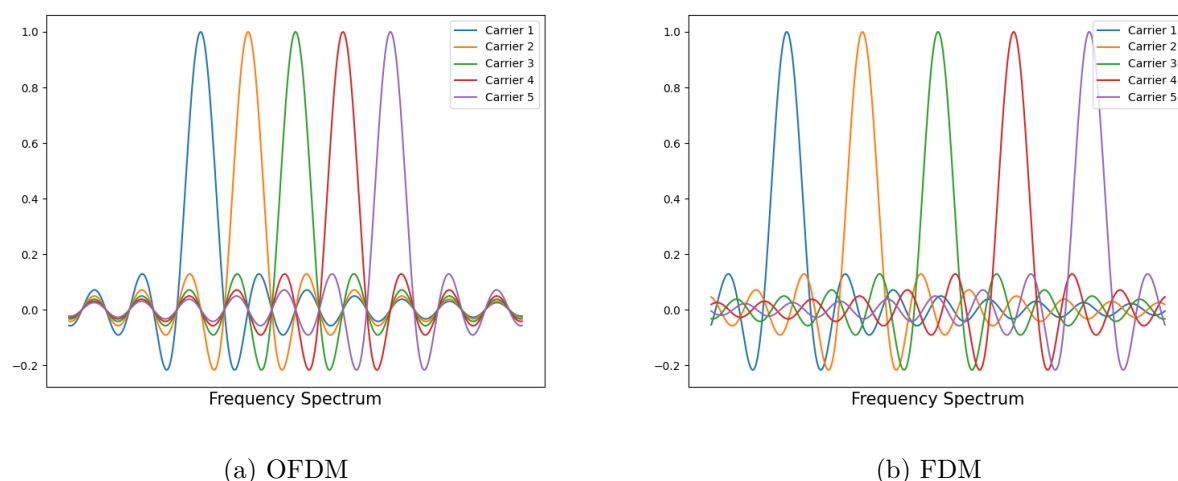


Figure 1.1: OFDM and FDM Waveforms

The above figure gives a broad idea of what goes into an OFDM-based signal, however in practice, the signal used will have vastly more complexity added on concerning error correction and data formats in general. The base signal however will still revolve around the idea of orthogonal carriers.

1.3 Fresnel Zones

1.4 LoRa Theory of Operation

1.4.1 Frequency Shift Keying

1.4.2 The Chirp

1.4.3 Bandwidth

1.4.4 Spreading Factor

1.4.5 LoRa Data Packets

1.4.6 LoRa Transmission Parameters

Symbol Rate and Symbol Time

Preamble Time

Payload Time

Packet Time

1.5 Critique of Literature

Bibliography

- [1] M. Skolnik. Radar - detection, military, technology | britannica. [Online]. Available: <https://www.britannica.com/technology/radar/History-of-radar>
- [2] W. Christian. Physical fundamentals of the radar principle. [Online]. Available: <https://www.radartutorial.eu/01.basics/Physical%20fundamentals%20of%20the%20radar%20principle.en.html>
- [3] S. B. Weinstein, “The history of orthogonal frequency-division multiplexing.” [Online]. Available: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=5307460>
- [4] G. Wright. What is orthogonal frequency-division multiplexing (OFDM)? [Online]. Available: <https://www.techtarget.com/searchnetworking/definition/orthogonal-frequency-division-multiplexing>

Glossary

AT&T American Telephone and Telegraph Company

FDM Frequency Division Multiplexing

OFDM Orthogonal Frequency Division Multiplexing

TDM Time Division Multiplexing

TDMA Time Division Multiple Access